Mohammad Hossein Hashemi

AI Engineer & Python Developer

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Profile

Machine Learning Engineer with 5+ years of experience in **deep learning, computer vision, and backend development**. Experienced in building **scalable AI solutions** and **robust RESTful APIs**. Passionate about **innovation, delivering high-impact projects**, and **continuous learning**. Interested in advancing expertise in **Computer Vision, Reinforcement Learning**, and **Large Language Models (LLMs)**.

Work Experience

AI Engineer | 2025-06– 2026-01 | Remote (Freelancer)

- Developed **real-time quantitative forecasting systems** for **BTC, SOL, USDT, and XAU**, integrating live market data via **Binance API** and **WebSocket streams** for low-latency processing.
- Implemented **multi-horizon forecasting** (5-min, 1-hour, 1-day, 60-step ahead) using ensemble models including **LSTM, GRU, Transformer, N-BEATS, Temporal Fusion Transformer (TFT), Prophet, ARIMA, XGBoost, LightGBM, CatBoost**, achieving **~45% RMSE reduction** and **R^2 up to 0.97**.
- Applied **feature engineering and preprocessing pipelines** using **Pandas, NumPy, TA-Lib, scikit-learn, and PyTorch**, including lag features, technical indicators, and rolling window statistics.
- Optimized **inference pipelines** with **CUDA, PyTorch JIT, ONNX, and TensorRT**, reducing decision latency by **~30%** and supporting live trading deployment.
- Built **real-time model evaluation dashboards** with **Weights & Biases, Matplotlib, Seaborn, and Plotly** for continuous monitoring of performance and drift detection.
- Implemented **automated retraining, hyperparameter tuning, and experiment tracking** with **Optuna, MLflow, and Weights & Biases** for reproducibility and continuous improvement.
- Developed high-performance **C++ optimization algorithms** for the **Maximum Clique Problem** using **Genetic Algorithms, Tabu Search, Simulated Annealing, and Reinforcement Learning**.
- Created **Python bindings** with **PyBind11** to integrate C++ algorithms into Python workflows for rapid experimentation and prototyping.
- Built **automated benchmarking pipelines** with **PyTest**, achieving **+38% improvement** in solution quality over baseline heuristics.
- Implemented **CI/CD pipelines** using **GitHub Actions and Docker**, ensuring reproducible builds, automated testing, and production-ready code quality.

AI Engineer | 2023-11 - 2025-06| Remote (Rayan Pezhohan)

- Architected and deployed a **scalable, real-time LLM platform** for real estate applications using **OpenAI GPT-3.5/4, LangChain, Haystack**, and locally hosted models (**LLaMA, Falcon**), supporting structured queries, rule-based logic, and multi-use workflows.
- Implemented **Retrieval-Augmented Generation (RAG)** with **FAISS** and **Milvus** vector databases, reducing hallucinations by **~45%** and improving factual accuracy in production.
- Designed **low-latency inference pipelines** with optimized prompting, context management, response validation, and batching, enabling stable concurrent request handling at scale.
- Built and maintained a **containerized backend architecture** with **Docker, Django, FastAPI, and PostgreSQL**, exposing RESTful APIs for real-time inference, session orchestration, and user-state management.
- Evaluated, benchmarked, and integrated **cloud and local LLMs** (GPT-3-class, LLaMA 2, Falcon), implementing model selection, fallback strategies, and hybrid routing to balance cost, latency, and reliability.
- Applied **caching, retrieval reuse, and API optimizations** using **Redis, Celery, and RabbitMQ**, achieving **~30% reduction** in end-to-end response latency under production workloads.
- Implemented **secure authentication and multi-tenant session management** with **JWT, OAuth2, and role-based access control**, supporting enterprise-grade deployments.
- Designed and trained **YOLOv8 models** for license plate detection (**95% accuracy**) and character recognition (**92.5% accuracy**) using **PyTorch (Ultralytics), OpenCV, Albumentations, and TorchVision**.
- Optimized **training and inference pipelines** with **CUDA, cuDNN, and mixed-precision training (AMP)**, reducing training time by **~60%** and achieving **<160 ms inference per frame**.
- Built and maintained backend infrastructure using **Flask, Python asyncio, and socket programming**, supporting **100+ concurrent users** with <50 ms response times.
- Leveraged **MLflow** for experiment tracking, model versioning, and performance monitoring, ensuring reproducibility across multiple iterations.

Computer Vision Engineer | Arooj | 2023-12 – 2024-11 | On-Site

- Built and optimized a **Hand Vein Detection Device** using **deep learning models** (**U-Net (89% IoU), SegNet, YOLOv5**) on **Raspberry Pi**, leveraging **PyTorch, TensorFlow, ONNX, TensorRT**, and **OpenCV** for real-time inference acceleration.
- Developed a **Shrimp Head Cutter System** using **OpenCV, NumPy, scikit-image**, and image processing techniques (**histogram equalization, edge detection, morphological operations**), achieving >95% real-time accuracy.
- Reduced **model inference time by 30%** through **quantization, pruning, batch optimization, and multi-threaded processing**, maintaining high performance on edge devices.

Research & Academic Experience

AI Researcher | 2024-07 – Present | Liverpool John Moores University | Remote(Part-Time)

- **AI & Machine Learning Research:** Developed and led **deep learning and machine learning projects** in **healthcare (infant cry analysis), animal behaviour, and intelligent systems**, demonstrating expertise in **signal processing, acoustic and visual feature extraction, model optimisation, and experimental evaluation**.
- **Scientific Publication Management**: **Managed the preparation and submission of research articles**, ensuring **clarity, scientific rigor, and fluency**. **Published a Q1 journal paper** and currently **submitting 7 additional Q1 papers**, highlighting strong research productivity and impact.

AI & Python Mentor | 2025-08 – Presents | Persian Gulf University | Remote(Part-Time))

- **AI & Python Instruction:** Taught **Python programming and AI concepts**, providing structured learning and **hands-on exercises** to build strong technical foundations for students and junior researchers.
- **Mentorship & Project Leadership:** Mentored **students and project teams** in designing and developing **AI-based applications and backend systems**, guiding projects from **initial concept to deployment** while fostering practical skills and teamwork.

Publications

Infant Cry Analysis: A Survey of Datasets, Features, and Machine Learning Techniques

Projects

Baby Crying Translator Mobile Application -Image to PDF - Tomato Leaf Disease Detection - Cat Voice Translator - ADHD Diagnosis

Education (2018 – 2023)

Bachelor of Electrical Engineering (Telecommunications) – Persian Gulf University

Skills

Python, MATLAB, C, C++, JavaScript, Machine Learning, Deep Learning, Computer Vision, NLP, Data Science, Backend Development, REST APIs, FastAPI, Flask, Django, Docker, Nginx, AWS, PostgreSQL, Redis, MongoDB, Git, Linux, PyTorch, TensorFlow, OpenCV, PyBind11, CI/CD, GitHub Actions, Reinforcement Learning, LLMs, LangChain